YIYAO ZHU

Postdoctoral Research Fellow, Department of Economics, National University of Singapore

yiyaoz@nus.edu.sg | yiyao-zhu.github.io

Academic Appointments

National University of Singapore, Department of Economics
Postdoctoral Research Fellow
Mentor: Professor Yingkai Li
2025–Present

Education

National University of Singapore
Ph.D. in Economics
Dissertation: *Essays on Contests*
Advisor: Professor Jingfeng Lu
2020–2025

National University of Singapore
B.Sc. (Hons.) in Applied Mathematics
Specialization: Operations Research and Financial Mathematics
Second Major: Economics
2016–2020

Research Papers

Optimal Public Programs: Transfers versus Ordeals *Working Paper*
with Yingkai Li

Abstract. We study the optimal design of public programs when scarce goods are allocated using either monetary transfers or costly ordeals. With risk-averse agents, the two screening instruments generate sharply different mechanisms. Under costly ordeals, an optimal mechanism takes a simple two-ticket form: intermediate types receive a rationed standard option, while high types obtain guaranteed service at a higher cost. Under monetary transfers, allocation probabilities vary more finely with type, with payments characterized by a common price wedge and endogenous entrance fees.

Prizes and Delegation in Team Competitions *Working Paper*
with Hideo Konishi and Jingfeng Lu

Abstract. This paper studies the design and delegation of prizes in competitions among teams whose heterogeneous members jointly produce output and whose noisy performance determines the winner. Our central result is a decentralization theorem: when a designer commits only to team-level winning and losing budgets and each coach freely allocates prizes within her own team to maximize that team's chance of winning, the equilibrium implements the efficient effort profile and extracts the available surplus. The instruments are outcome-contingent individual prizes: the win–lose spread aligns marginal incentives, while losing-prize levels extract rents. These results rest on a reduction that represents any within-team sharing rule by a single capability index, turning the team game into an asymmetric Tullock contest and characterizing the sharing rules that maximize a team's winning probability or aggregate output.

Dividing the Spoils: Strategic Prize Allocation in Team Contests *Under Review*
with Zhonghong Kuang and Jingfeng Lu

Abstract. Rival teams compete through battles and reward members only after team victory. We study how team managers allocate victory-contingent reward budgets across players to maximize their teams' winning probabilities in a majoritarian contest. Under a regularity condition on battle technologies, the allocation game has a unique pure-strategy equilibrium. Despite differences in budgets and player costs across teams and technologies across battles, both managers choose the same normalized reward schedule, a property we call reward schedule alignment. Each battle's share is proportional to discriminatory power, closeness, and pivotality. With precommitted rewards, allocations and team-winning probabilities are independent of battle order.

Move Orders in Contests: Optimal Handicaps and Effort Ranking *Working Paper*

with Lei Gao, Jingfeng Lu, and Zhewei Wang

Abstract. We study how a contest designer should jointly choose identity-contingent favoritism and move order in a two-player generalized Tullock contest to maximize expected total effort. We show that, when favoritism can be chosen freely, weak-player leadership is never optimal under cost asymmetry. When contest accuracy is relatively low, strong-player leadership is optimal. If the players are sufficiently similar in ability, the designer favors the strong leader to reinforce its commitment incentive; if the ability gap is sufficiently large, the designer instead favors the weak follower to level the playing field and preserve competitive pressure. When contest accuracy is sufficiently high, the optimal strong-lead contest is preemptive and, under cost asymmetry, favors the weak follower, thereby forcing the strong leader to exert more effort to deter entry. Simultaneous play can be optimal only when contest accuracy is high and the players are extremely similar in ability.

Where Teams Randomize: Multi-Task Team All-Pay Contests *Working Paper*

with Jingfeng Lu

Abstract. We study a complete-information all-pay contest between two teams performing $K \geq 2$ complementary tasks under Cobb–Douglas aggregation, and we completely classify *one-pair mixed* equilibria, in which exactly one member of each team randomizes. Randomization can occur only on maximal-weight tasks: a same-task pair requires maximal weight, a cross-task pair requires both tasks maximal, and each labeled pair supports one equilibrium, yielding exactly $|\mathcal{K}^*|^2$ equilibria, where $\mathcal{K}^*$ is the set of maximal-weight tasks. All of them induce identical member-by-member payoffs, expected costs, and winning probabilities, governed by a single effective cost ratio, and all weakly Pareto-dominate the all-zero equilibrium. Each equilibrium path survives block-sequential timing exactly when its randomizers move in the terminal block.

Seminars and Conferences

2025 NUS Micro Theory Lunches (Singapore)

2024 NUS Micro Theory Lunches (Singapore); Asian Meeting of the Econometric Society (Hangzhou); Conference on Mechanism and Institution Design (Budapest); Asian Meeting of the Econometric Society (Ho Chi Minh City); East Asia Game Theory Conference (Jeju); China Meeting on Game Theory and Its Applications (Shenzhen); GAMES 2024 (Beijing)

2023 Asian Meeting of the Econometric Society (Singapore)

Awards and Fellowships

Research Scholarship, Ministry of Education, Singapore
2023–2024

Research Scholarship, Department of Economics, National University of Singapore
2024–2025

Graduate Teaching Fellowship, National University of Singapore
2020–2023

Science and Technology Undergraduate Scholarship, National University of Singapore
2016–2020

Teaching Experience

EC3102 Macroeconomic Analysis II, Teaching Assistant, National University of Singapore
2022–2024

Professional Service

Referee: *International Journal of Game Theory*; *The Japanese Economic Review*